

SONiX 32-Bit Cortex-M0 Micro-Controller

DMA UART Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/27	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of UART in DMA mode.

2 FEATURE

- UART DMA TX Mode.
- UART DMA RX Mode.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect UTXD0 and URXD0.

4 EXAMPLE CODE

4.1 Initialization

Initial UART and DMA function. User can modify the settings of UART and DMA in these function.

UART TX Mode : DMA channel would move data from other peripherals or memory to UART transmitter holding register when UART THR is empty.

UART RX Mode : DMA Channel would move data from UART receive buffer register to other peripherals or SRAM when RX data buffer has data.

```

78 // Initial UART
79 UART0_Init();
80
81 // Disable UART ISR when using DMA UART
82 NVIC_DisableIRQ(UART0_IRQn);
83
84 // Init DMA
85 DMAUart0TXInit();
86 DMAUart0RXInit();

```

<pre> 124 void DMAUart0TXInit(void) 125 { 126 DMA_InitSt stdMACH_Init; 127 128 // Set DMA CH0 as M(SRAM) to P(UART TX) mode 129 // Source : Memory, Increment, 8bit, no request, Burst 1 130 stdMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_INC; 131 stdMACH_Init.b_SrcMode = DMA_SRC_MEMORY; 132 stdMACH_Init.b_SrcWidth = DMA_SRC_WIDTH_8BIT; 133 stdMACH_Init.b_SrcReqSel = DMA_SRCRS_NONE; 134 stdMACH_Init.b_BurstSize = DMA_BURST_1; 135 // Destination : Peripheral UART, Fixed, 8bit, UART0 TX 136 stdMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_FIX; 137 stdMACH_Init.b_DstMode = DMA_DST_PERIPHERAL; 138 stdMACH_Init.b_DstWidth = DMA_DST_WIDTH_8BIT; 139 stdMACH_Init.b_DstReqSel = DMA_DSTRS_UART0_TX; 140 // Channel : LV0, Enable TC/ABT interrupt 141 stdMACH_Init.b_Priority = DMA_CHPRI_LV0; 142 stdMACH_Init.b_IntTCEn = DMA_INT_TC_MSK_DIS; 143 stdMACH_Init.b_IntABTEn = DMA_INT_ABT_MSK_DIS; 144 145 // Initial DMA 146 DMA_Init(&stdMACH_Init, eDMA_CH0); 147 } </pre>	<pre> 157 void DMAUart0RXInit(void) 158 { 159 DMA_InitSt stdMACH_Init; 160 161 // Set DMA CH1 as P(UART RX) to M(SRAM) mode 162 // Source : Peripheral UART, Fixed, 8bit, UART0 RX, Burst 1 163 stdMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_FIX; 164 stdMACH_Init.b_SrcMode = DMA_SRC_PERIPHERAL; 165 stdMACH_Init.b_SrcWidth = DMA_SRC_WIDTH_8BIT; 166 stdMACH_Init.b_SrcReqSel = DMA_SRCRS_UART0_RX; 167 stdMACH_Init.b_BurstSize = DMA_BURST_1; 168 // Destination : Memory, Increment, 8bit, no request 169 stdMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_INC; 170 stdMACH_Init.b_DstMode = DMA_DST_MEMORY; 171 stdMACH_Init.b_DstWidth = DMA_DST_WIDTH_8BIT; 172 stdMACH_Init.b_DstReqSel = DMA_DSTRS_NONE; 173 // Channel : LV0, Enable TC/ABT interrupt 174 stdMACH_Init.b_Priority = DMA_CHPRI_LV0; 175 stdMACH_Init.b_IntTCEn = DMA_INT_TC_MSK_DIS; 176 stdMACH_Init.b_IntABTEn = DMA_INT_ABT_MSK_DIS; 177 178 // Initial DMA 179 DMA_Init(&stdMACH_Init, eDMA_CH1); 180 } </pre>
--	--

UART will transmit 256 bytes of data from 0x00 to 0xFF.

```

88 // Init data buffer
89 for(i = 0; i < DMA_TEST_SIZE; i++)
90 {
91     // Set Tx data = 0x00, 0x01, ... , 0xFF
92     b_aryUartTXData[i] = i & 0xFF;
93 }

```

4.2 DMA UART function enable

UART DMA RX mode must be enabled before UART TX is started to avoid missing data on UART bus.

```
95 // Start DMA UART RX transfer before UART TX
96 UART0_DMAEnable();
97 UART0_DMASizeSet(DMA_TEST_SIZE);
98 DMAUart0RXStart();
99 // Start DMA UART TX transfer
100 DMAUart0TXStart();
```

The DMA TC flag and interrupt will be set when all data transfers are completed.

```
102 // Wait for DMA TC interrupt flag
103 while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);
104 while(stDMA0IntFlag[eDMA_CH1].Flag.bits.TC != 1);
```

5 RESULT

The data(0x00~0xFF) sent or received can be observed on UART bus.



The results of DMA can be observed through Watch 1 window or Memory window.

Memory 1	Memory 2
Address: b_aryUartTXData	Address: b_aryUartRXData
0x2000001C: 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F	0x2000001C: 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
0x2000002C: 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F	0x2000002C: 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
0x2000003C: 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F	0x2000003C: 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F
0x2000004C: 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F	0x2000004C: 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F
0x2000005C: 40 41 42 43 44 45 46 47 48 49 4A 4B 4C 4D 4E 4F	0x2000005C: 40 41 42 43 44 45 46 47 48 49 4A 4B 4C 4D 4E 4F
0x2000006C: 50 51 52 53 54 55 56 57 58 59 5A 5B 5C 5D 5E 5F	0x2000006C: 50 51 52 53 54 55 56 57 58 59 5A 5B 5C 5D 5E 5F
0x2000007C: 60 61 62 63 64 65 66 67 68 69 6A 6B 6C 6D 6E 6F	0x2000007C: 60 61 62 63 64 65 66 67 68 69 6A 6B 6C 6D 6E 6F
0x2000008C: 70 71 72 73 74 75 76 77 78 79 7A 7B 7C 7D 7E 7F	0x2000008C: 70 71 72 73 74 75 76 77 78 79 7A 7B 7C 7D 7E 7F
0x2000009C: 80 81 82 83 84 85 86 87 88 89 8A 8B 8C 8D 8E 8F	0x2000009C: 80 81 82 83 84 85 86 87 88 89 8A 8B 8C 8D 8E 8F
0x200000AC: 90 91 92 93 94 95 96 97 98 99 9A 9B 9C 9D 9E 9F	0x200000AC: 90 91 92 93 94 95 96 97 98 99 9A 9B 9C 9D 9E 9F
0x200000BC: A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF	0x200000BC: A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF
0x200000CC: B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD BE BF	0x200000CC: B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD BE BF
0x200000DC: C0 C1 C2 C3 C4 C5 C6 C7 C8 C9 CA CB CC CD CE CF	0x200000DC: C0 C1 C2 C3 C4 C5 C6 C7 C8 C9 CA CB CC CD CE CF
0x200000EC: D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 DA DB DC DD DE DF	0x200000EC: D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 DA DB DC DD DE DF
0x200000FC: E0 E1 E2 E3 E4 E5 E6 E7 E8 E9 EA EB EC ED EE EF	0x200000FC: E0 E1 E2 E3 E4 E5 E6 E7 E8 E9 EA EB EC ED EE EF
0x2000010C: F0 F1 F2 F3 F4 F5 F6 F7 F8 F9 FA FB FC FD FE FF	0x2000020C: F0 F1 F2 F3 F4 F5 F6 F7 F8 F9 FA FB FC FD FE FF

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